Work and Energy[1]

Disha Kuzhively

April 30, 2026

[1] Draft version as of April 30, 2026

This article was prepared by the author to serve as a draft for TNSCERT class 7 chapter on the concept of Work and Energy during the the Tamil Nadu Curriculum Framing (TNCF) workshop that was held in Asha Nivas, Chennai from 13-11-2025 to 15-11-2025, 09-03-2026 to 10-03-2026, and 27-04-2026 to 30-04-2026.

The right side margin of each page contains notes for teachers, and prompts to activities that the teacher can initiate in the classroom.

Draft Syllabus[2]:

C 1.11: Understands and calculates work and energy, identifies different forms and sources of energy, observes and represents energy changes in real-life situations.

Work and Energy: Work: Concept and definition of work in science - quantification of work done by a steady force, Simple numerical problems, SI Unit of Work (Joule)

C 1.12: Applies scientific knowledge and uses appropriate scientific vocabulary to communicate ideas and to promote responsible use of energy.

Energy: Concept and definition of energy, different forms of energy, conversion of energy, energy flow in daily life systems, useful and wasted forms of energy, law of conservation of energy, sources of energy - renewable and non-renewable energy sources-conservation

[2] TNSCERT. Tamil nadu curriculum framework 2025 - proposed draft syllabus, pedagogical approaches and assessment strategies - classes 6- 8 - science. 2025

In earlier lessons, we learned that a force is needed to start motion, stop motion, change speed, or change the direction of motion.

Let us do a simple activity. Place a book on the desk and push it slowly so that it slides at a steady speed. By pushing slowly, we are trying to apply a constant force to move the book. In everyday language, we say that we are doing some *work*. If we push the same book a longer distance or if we are pushing a heavier stack of books the same distance, we are doing more work than we did initially.

Illustration of a hand sliding a book across a table.

Illustration of a hand sliding a book across a table for a long distance.

Illustration of a hand sliding a stack of books across a table.

Figure 1: Sliding a book across a table.

From this, we can see that the work done depends on two things:

1. how much force is applied, and
2. how far the object moves.

When a constant force (F) moves an object through some dsitance (d) in the same direction as the force, the work done (W) is defined as:

$$W = F \times d \tag{1}$$

The SI unit of work is Joule (J), named for James Prescott Joule, an English scientist of the 1800s.

The SI unit of force is Newton (N) and the SI unit of distance is metre (m). This makes

$$1\ \mathrm{J} = 1\ \mathrm{N} \times 1\ \mathrm{m} = 1\ \mathrm{kgm/s^2} \times 1\ \mathrm{m} = 1\ \mathrm{kgm^2/s^2} \tag{2}$$

EXAMPLE: If we apply a force of 10 N to push the book a distance of 2 m, the work done = 10 N × 2 m = 20 J.

QUESTION: In which of the following cases is work being done? In which of these cases are we applying force but there is no movement?

1. Holding a heavy bag
2. Pushing a wall
3. Climbing stairs
4. Kicking a ball
5. Sitting and reading

Remember that we are not concerned here with the common meaning of the word "work," which implies that any physical or mental labor is work. We are only concerned with work where there is a force acting on an object and the object moves in the direction of the force.

QUESTION: Why is pushing a wall not considered work in physics?

When is work done zero?

From the definition of work done (Equation 1), we can see that work done is zero if no force is applied by the agent , i.e., $F = 0$. If the object does not move, $d = 0$, then no work done even if you are applying a force. For example, if you push a wall and it does not move, the work done is zero.

Work is done only when the force acts in the same direction as the movement. If the force acts in a direction perpendicular (at a right angle) to the movement, then the work done by that force is zero.

EXAMPLE: Thulasi pushes a book across the table, and the book moves forward. At the same time, Vaishali presses the book downward into the table. The book does not move downward. Thulasi is doing work because her force causes movement. Vaishali is not doing work because there is no movement in the direction of her force. So, the work done by Vaishali is zero.

Illustration of a hand pushing a book into the table while another hand is pushing a book across the table.

Figure 2: Pushing a book into the table

Can work done be negative?

Think about pushing a book across a table. The book moves forward because we apply a force. At the same time, the table exerts a friction force on the book in the opposite direction.

Remember that in formula for work (Equation 1), distance (d) is measured in the direction of the force. In this situation the book moves forward and the frictional force acts backward. So, the motion of the book is in the opposite direction to the friction force. This means the distance, when measured in the direction of the friction force, is negative. Therefore, the work done by friction is negative.

If the force and the motion are in opposite directions, the work done is negative. Negative work does not mean "no work." It means that the force is acting against the motion. It tries to slow the object down or stop it.

QUESTION: We apply brakes to slow down and stop a cycle. The cycle is moving forward, but the braking force acts backward. Is the work done by the braking force positive or negative?

Energy

Energy is the capacity to do work. Let us understand this with an example. Imagine a stack of books kept on the floor. Let us pick up some of the books and place them on a table. To do this, we apply an upward force and move the books through a certain height. So, we have done work.

When we do work on the books, we are also transferring energy to them. The books on the floor have less energy than the books on the table that are at a higher position.

Similarly, imagine a football resting on the ground. It is not moving. When we kick the ball, we are applying a force to it over a short distance. So, we do work on the ball. As a result, energy is transferred from our body to the ball. This energy makes the ball move forward.

If we kick the ball harder, we are applying a larger force. So, we are doing more work, and more energy is transferred to the ball. The ball then moves faster and travels a longer distance. If the ball slows down and stops after some time, it is because forces like friction act on it and remove energy from it. In this way, by kicking the ball, we can clearly see how doing work transfers energy and causes motion.

Relationship between work and energy

Work and energy are closely related. When we do work on an object, we transfer energy to it. When an object does work on something else, it transfers energy from itself. Energy transferred to the object is positive work, and energy transferred from the object is negative work.

Work done on an object, its energy changes. We can say that work done on an object is the change in its energy. This means, if positive work is done, the energy of the object increases and if negative work is done, the energy of the object decreases. This idea is true even when the force is not constant or changes during motion.

The SI unit of both work and energy is the (J).

Our body gets the energy from the food we consume. The energy stored in food is conventionally measured in the unit of calorie. 1 cal (calorie) = 4.2 J.

EXERCISE: Check the labels of packaged food items in your house and list the calories contained in each food item.

James Prescott Joule studied how mechanical energy and thermal energy are related, and can be converted from one to the other. This helped develop a unified way to understand energy. Conducted fundamental researches on the properties of heat and formulated the equation related to the heat produced when electricity flows. Played an important role in formulating the theory of thermodynamics by connecting mechanical energy and heat energy. In 1850 he was selected as a fellow of the Royal Society.

Image of James Prescott Joule.

Figure 3: James Prescott Joule

We require energy to do a lot of everyday tasks. For example, in a classroom, At home heating milk to drink, all of this work requires energy.

Strait of Hormutz Energy crisis Efficient use of energy

Energy is associated with state (of motion, position) temperature (motion of molecule) do we mention this? shall we restrict to kinetic energy only?

conserved

can be changed from one form to another (like money)

kinetic energy

potential energy, while cycling uphill is difficult, need to put more energy to peddle

Roller coaster during fairs, how do they work?

Suddenly you realize that a car is headed toward your car the wrong way in a one-way tunnel. To minimize your danger from the impending accident, should you match your speed to that of the other car, go even faster, or slow to a stop? A head-on collision is the most dangerous type of car col- lision. Surprisingly, the data collected about head-on collisions suggest that the risk (or probability) of fatality to a driver is less if that driver has a passenger in the car. Why is that?

The best advice is to stop and, if possible, put your car in reverse. You can get a measure of the severity of the collision by considering the total kinetic energy or the total momentum of the cars before the collision. If you do not decrease your speed toward the other car, both quanti- ties are large, and so the collision will be severe. The situation is unlike football, where one player may choose to speed up when running head-on into another player. The difference is that a player may want the colli- sion to be violent, and by properly orienting his body he can shift the collision to his opponent's vulnerable area or cause his opponent to become unbalanced and slam into the field. Data collected about head-on car collisions indicate that adding a passenger to your car reduces your risk of fatality. That risk depends on the change in your velocity during the collision: A large change means that you undergo a severe acceleration due to a severe force. For example, if your car has a small mass and the other car has a large mass, your veloci- ty may be changed so much that you end up going backward. Additional mass in your car, from a passenger or even a sand- bag in the trunk, can decrease your change in velocity and thus also your risk. Here is one numerical result: Suppose your car and the other car are identical and that your mass and the other driver's mass are identical. Your fatality risk is reduced by about 9in your car.

examples from cricket

newton's cradle

Why do balls bounce?

Rock climbers (as opposed to spelunkers) use rope that gives considerably under stress, so that if you fall, your stop at the end of the fall is not sudden and the force stop- ping you is not huge. As the rope begins to stretch, the rope materials rub against one another and become warmer; most of the potential energy and kinetic energy that you lose dur- ing the fall ends up as thermal energy within the rope.

Playing carroms

Working of a yoyo - when talking about types of mechanical energy

Playing using spinning tops

Energy as "ability to do work"**

Goal

Students see energy as a unifying idea across situations and begin identifying forms.

—

Warm-up (10 min): Recall + puzzle

Ask:

> Yesterday, when work was done, what made it possible?

Let them say:

* "we used energy" (even if vague)

—

Activity 1 (25 min): "What caused the change?"

Stations (rotate groups):

1. Stretched rubber band → released 2. Ball at height → dropped 3. Rolling ball → hits another 4. Battery + bulb circuit 5. Rubbing hands → heat

Worksheet table

| Situation | What changed? | What caused it? | Where was it before? | | ——— | ———- | ————— | ——————— |

Goal: They notice "stored" vs "moving" vs "transfer"

—

Discussion (15 min): Naming carefully

Introduce:

> Energy = capacity to do work

Then label (only after observation):

* Motion → kinetic energy * Height/position → potential energy * Stored in stretch → elastic energy

Avoid listing beyond these 3.

—

Activity 2 (20 min): Comparison tasks

Ask:

* Which has more energy?

* Fast vs slow ball * Heavy vs light ball * High vs low object

Students justify qualitatively.

Then introduce:

$"math_block_widget_always_prefetch_v2" : "content" : "KE = \frac{1}{2}mv^2"$

Use it only for:

* "What changes more: doubling speed or doubling mass?"

Let them test with small numbers.

—

Activity 3 (15 min): Energy chains

Give real-life cases:

* Eating → running * Sun → drying clothes * Battery → torch

Task: Draw arrows Example:

> Food → muscle → motion → heat

—

Exit question

> Where does the energy go when a moving object stops?

—

Day 3: Energy change, representation, and responsibility

Goal

Students represent energy transformations and connect to everyday decisions.

—

Warm-up (10 min): Challenge

> A ball rolls, slows down, and stops. > Has energy disappeared?

Let them argue.

—

Activity 1 (25 min): Tracking energy

Situations

1. Falling object 2. Cycling and braking 3. Electric fan 4. Boiling water

Task (worksheet)

Fill:

Step	What happens	Energy form before	Energy form after

—

Discussion (15 min): Big idea

Guide toward:

> Energy changes form, it does not just vanish.

(No need to formalise conservation law explicitly.)

—

Activity 2 (20 min): Design challenge
> "Design a system that uses energy usefully"
Examples:
* Solar cooker * Efficient classroom lighting * Insulated water bottle
Constraints:
* Identify source of energy * Show transformations * Explain how waste is reduced
—
Activity 3 (15 min): Language upgrade
Give sentences:
* "Energy is lost" * "Work is done by pushing hard" * "Fan uses electricity"
Task: Rewrite using scientific vocabulary
Example:
> "Electrical energy is converted into kinetic energy of the fan."
—
Final reflection
Short writing:
> One situation where understanding energy can help reduce waste.
—
Worksheet Pack (summary)
Sheet 1: What is work?
* Yes/No classification * Explain contradictions * Identify role of displacement
Sheet 2: Observing energy
* Table of situations * "What caused the change?"
Sheet 3: Comparing energy
* Qualitative comparisons * Simple KE calculations
Sheet 4: Energy chains
* Arrow diagrams
Sheet 5: Energy tracking
* Before/after tables
—
Pedagogical notes (important for your style)
* Let disagreement sit longer than comfortable * Do not correct immediately; redirect with examples * Use measurement only to support thinking, not dominate it * Keep formal vocabulary minimal but precise * Revisit earlier examples after each new idea (spiral, not linear)
—
If you want, I can convert this into **printable worksheets (LaTeX-ready, clean formatting)** or add **assessment rubrics aligned with competencies**.

EXERCISE: Write down all the work that you do in your daily life.

Where do these sounds come from? How are these sounds produced? In this chapter, we will explore how sound is produced and how it propagates[3] from one place to another. We will also learn that sound can be understood as a wave.

[3] Here the word 'propagate' is used instead of simpler and familiar word like 'travel' to avoid misconceptions that students create, e.g. The word travel suggests that 'sound' is a moving object. Words are chosen to ensure that students develop a scientifically accurate mental model of the concept.

How is Sound Produced?

Activity: Vibrating Scale

Materials required:

1. Scale
2. Desk or table

Place the scale on the desk so that about one-third of it is over the edge. Hold the part on the desk with one hand. Push the free end of the scale downward. Release it quickly.

OBSERVE: do you see ripples forming on the surface of the water?

DISCUSSION: In each case, some part of the object moves back and forth rapidly. This type of motion is called **vibration**.

Did you know? Many quartz clocks and watches keep time using a small quartz crystal. This crystal is shaped like a tiny tuning fork.

SOUND IS PRODUCED WHEN OBJECTS VIBRATE. The vibrating part of an object can make nearby air move. These vibrations spread through the air and reach our ears as sound.

EXERCISE: create a table as shown in Table listing the object that produces sound and the part of the object that vibrates.

Sl. No.	Object that produces sound	Part of the object that vibrates
1	Vibrating scale	Tip of the scale
2	Musical geometry box	⋮
3	⋮	⋮
⋮	⋮	⋮

Table 1: Observation table

Speak into one cup while your benchmate listens through the other cup. See Figure **??**

Observe: Can your benchmate hear the voice? What happens to the sound when you let the string loose?

Discussion: When the string is tight, vibrations travel through the string. When the string is loose, the vibrations do not travel well. This shows that sound needs a material medium to propagate.
called a **medium**.

Observation No.	No. of times the scale moves back and forth in 5 seconds (n)	$n/5$
1	...	...
2	...	...
3	...	...

Table 2: Observation table for finding the frequency of the vibrating scale

When the scale vibrates, it also makes the surrounding air vibrate. As the scale moves back and forth, it pushes and pulls the nearby air. This creates compressions and rarefactions in the air.

Discussion: the number of vibrations that occur in one second is called the **frequency** of the vibration. Frequency is measured in hertz (Hz) and usually represented by the Greek letter ν (nu). Frequency tells us how often an event occurs in a unit of time.

Example: if you beat a drum 10 times in one second, the frequency of your beating is 10 Hz.

Pitch

Sounds with higher frequency have a higher pitch and sounds with lower frequency have a lower pitch. A mosquito makes a high-pitched sound. A drum makes a low-pitched sound. Which one is vibrating faster?

Question: A bell is struck and produces sound. After a few seconds, the sound stops. What must have happened to the bell?

Question: Kavin and Priya are standing at different distances from the lightning. Both hear the thunder. Will the pitch of the thunder be different for them? Will the loudness of the thunder be different?

Amplitude

You may also have noticed something else during the activity. When the scale was plucked gently, the sound was soft. When the scale was plucked harder, the sound was louder.

If you watch the scale carefully, you will see that when it is plucked harder, it moves farther away from its resting position during each vibration.

The size of this vibration is called the **amplitude**. Amplitude tells us how large the vibration is. A vibration with greater amplitude produces a louder sound, while a vibration with smaller amplitude produces a softer sound. Amplitude affects loudness, not pitch.

QUESTION: Two sounds have the same pitch, but one is louder. What must be different?

Pitch depends on frequency, which is determined by the source. Distance from the source does not change the frequency. As sound spreads out, its amplitude decreases, this is why it becomes quieter as you move far from the source.

Hearing loss and hearing aids

Very loud sounds can damage the inner ear and lead to hearing loss. Some people are born with conditions that affect their ability to hear.

Hearing aids are devices that help such people hear better. They work by amplifying sound, that is, by increasing the amplitude of the sound waves.

Activity: long spring

Take a long spring. Ask your classmate to hold one end while you hold the other end. Stretch the long spring gently between you. Now give the long spring a quick push toward your classmate.

OBSERVE: You will see regions where the coils are close together and regions where they are farther apart. A region where the coils are close together is called a compression. A region where the coils are farther apart is called a rarefaction. This is similar to what happens in sound waves in air.

Wavelength

The distance between two neighbouring compressions (or two neighbouring rarefactions) is called the **wavelength**. Wavelength is measured in the units of distance. The SI unit of distance is the metre (m). Wavelength is usually represented by the Greek letter λ (lambda). See Figure . Repeat the activity, but now create compressions more quickly.

OBSERVE: When vibrations happen more often each second (higher frequency), the compressions are packed closer together and the wavelength becomes shorter.

Speed of Sound

Do we hear sound instantly after it is produced? Think about what happens during lightning and thunder. When lightning strikes, we see a flash of light and hear thunder.

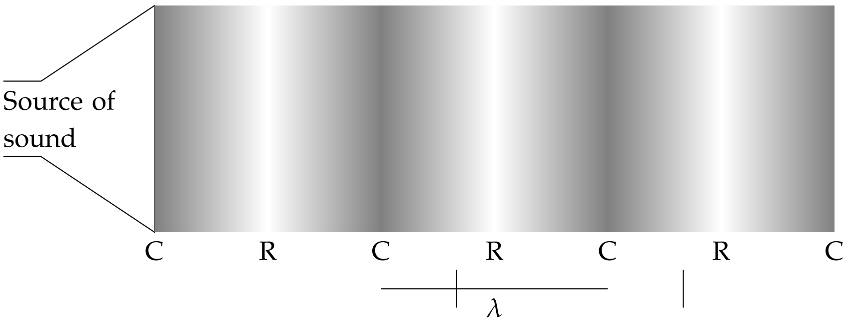

Figure 4: The distance between two neighbouring compressions (or two neighbouring rarefactions) is the wavelength (λ).

OBSERVE: Do you see the lightning and hear the thunder at the same time? Or do they happen one after the other? Which do you notice first?

Usually, we see the lightning first and hear the thunder later.

These observations show that sound takes time to propagate from the source to our ears. Light travels much faster than sound, so we see the flash before we hear the sound.

If the thunder sounds quieter when you are farther away, does the thunder have less energy when it reaches you?

In air, sound propagates at a speed of about 340 metres per second (at room temperature). The speed of a sound wave depends on both its frequency and its wavelength.

$$\text{Speed} = \text{Frequency} \times \text{Wavelength} \tag{3}$$

Sound does not propagate at the same speed in all materials. The speed of sound depends on the medium through which it propagates. Sound propagates fastest in solids, slower in liquids, and slowest in gases. For example, sound propagates faster in metal or wood than in air. This is because the particles in solids are closer together, allowing vibrations to pass more quickly from one particle to the next.

QUESTION: How does temperature affect the speed of sound?

As the temperature increases, the speed of sound increases. This happens because when the temperature is higher, the particles of the medium move faster. As a result, they pass the vibrations to neighbouring particles more quickly, allowing sound to propagate faster. For example, sound travels faster in warm air than in cold air.

Musical instruments

Musical instruments produce sound by making strings, membranes, or air columns vibrate. By changing certain parts of the instrument, musicians can change the frequency of vibration, and therefore the pitch of the sound.

Different instruments change the frequency in different ways.

Veena (string instrument): The strings vibrate to produce sound. Pressing the string at different positions changes the length of the vibrating string, which changes the frequency and the pitch.

Parai (drum): The stretched membrane (skin) vibrates when it is struck. Changing the tightness of the membrane changes the frequency of vibration and the sound produced.

Nadaswaram (wind instrument): Sound is produced by vibrations of the air column inside the instrument. Opening and closing the finger holes changes the effective length of the air column, which changes the frequency. Although these instruments look very different, they all produce sound through vibrations, and the frequency of vibration determines the pitch of the sound.

Ultrasound

Humans can hear sounds with frequencies roughly between 20 Hzand 20,000 Hz. Sounds with frequencies higher than 20,000 Hzare called ultrasound. Humans cannot hear ultrasound.

Ultrasound has many useful applications:

1. Medical imaging: Doctors use ultrasound to look inside the body, for example to observe a developing fetus during pregnancy.
2. Detecting cracks in materials: Ultrasound can be used to find cracks or defects in metals and other materials.
3. Animal communication and navigation: Some animals, such as bats and dolphins, produce and detect ultrasound. They use it to communicate and to find objects around them.

Echo

Sound travels at a finite speed. When a sound wave meets a surface such as a wall, cliff, or building, it can reflect and travel back toward the source.

An echo is heard when the reflected sound reaches our ears after a short delay.

For an echo to be heard clearly, the reflecting surface must be far enough away. If the surface is too close, the reflected sound mixes with the original sound and we do not hear a separate echo.

For example, when you shout in a large empty hall, valley, or near a cliff, you may hear your own voice repeated after a short time. This repeated sound is the echo. Echoes show that sound waves can reflect from surfaces and travel back to us.

Echolocation

Animals such as bats and dolphins produce sounds and listen to the echoes to locate objects around them.

Sonar

Ships and submarines use sound waves and their echoes to detect objects underwater and measure distances.

Noise Pollution

Unwanted or unpleasant sound is called noise. When there is too much noise in our surroundings, it leads to noise pollution.

Common sources of noise pollution include traffic noise from vehicles and horns, loudspeakers used at high volume,industrial noise from machines and factories, construction activities.

Continuous exposure to loud noise can cause stress, lack of sleep, and hearing problems. How can we reduce noise pollution?

We can reduce noise pollution by avoiding unnecessary honking on the roads, keeping the volume of loudspeakers low, using silencers and maintaining vehicles properly, planting trees, which can help absorb sound, following rules and regulations about noise levels.

Reducing noise pollution helps create a healthier and more peaceful environment.

If you pick up a seashell on a beach and hold it close to your ear, you may hear a sound that resembles waves breaking on the shore. Where does this sound come from?

Sound is an example of a mechanical wave. It is also a longitudinal wave, because the particles of the medium vibrate along the direction in which the wave propagates. There is another type of wave called a transverse wave. In a transverse wave, the particles of the medium move perpendicular to the direction in which the wave propagates. In other words, the particles move up and down, while the wave moves forward.

An example of a transverse wave can be seen when a pebble is dropped into a pond. The ripples that spread across the surface of the water are transverse waves.

Light is also a transverse wave. However, light does not need a material medium to propagate, so it is not a mechanical wave.

The world around us is full of waves - sound waves, water waves, and light waves.

References

David Halliday, Robert Resnick, and Jearl Walker. *Fundamentals of physics*. John Wiley & Sons, 2013.

Zdeslav Hrepic, Dean A. Zollman, and N. Sanjay Rebello. Identifying students' mental models of sound propagation: The role of conceptual blending in understanding conceptual change. *Phys. Rev. ST Phys. Educ. Res.*, 6:020114, Sep 2010. DOI: 10.1103/PhysRevSTPER.6.020114. URL https://link.aps.org/doi/10.1103/PhysRevSTPER.6.020114.

Henry Roscoe. *The Life and Experiences of Sir Henry Enfield Roscoe*. Macmillan: London and New York, 1906. URL https://commons.wikimedia.org/w/index.php?curid=3119134.

TNSCERT. Tamil nadu curriculum framework 2025 - proposed draft syllabus, pedagogical approaches and assessment strategies - classes 6- 8 - science. 2025.

Jearl Walker. *The flying circus of physics*. John Wiley & Sons, 2006.